

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Captive breeding, habitat creation & management - Set 1

Mark scheme

AQA spec 3.1.2.3.3-3.1.2.3.4; The living environment; textbook pp. 32-49

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching)	2
01.2	<i>Any 4 from:</i> • Is the wild population threatened? • Is there a genetically diverse captive population? • Is in-situ conservation failing? • Is keeping a captive population realistic (habitat size, food, relationships, cost)? • Is release likely to be successful now or in future (suitable habitat, threats removed)?	4
01.3	• Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
02.1	• Very small gene pool so individuals are closely related and inbreeding is likely • Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low • A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) • Exchange of individuals / semen between zoos; avoid hybridisation with related species / subspecies	4
02.2	<i>Any 3 from:</i> • Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided • Population interactions: mixing breeding and non-breeding individuals causes conflict; some need a choice of partners, others pair for life • Breeding habitat needed: courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) • Stress / disturbance	3
02.3	• AI: semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet; avoids dangerous mating (elephants, rhinos) and partner rejection • Embryo transfer: a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) • So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
03.1	• Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups	3
03.2	<i>Any 5 from:</i> • Large enough suitable habitat for a viable population • Reliable food supply • Low predation risk / original threat removed • Suitable breeding sites • Water • Support of the local human population • Official / legal protection of habitat and species	5
03.3	• Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed • Identifies causes of failure so future releases can be improved / extra support given	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	

Question	Marking guidance	Marks
04.1	• The nucleus of a stem cell from the endangered species is transferred into an empty egg of a closely related species (southern white rhino) • The embryo is implanted into a surrogate female of the related species which gives birth to a northern white rhino • Very low success rates with wildlife so far; deep-frozen cells could allow clones even after the last individuals die	3
04.2	• Micro-propagation: tissue culture producing many cell clusters from one plant, each grown into a genetically identical plant • Black poplar: cuttings / pollarded stems of pure native trees are rooted to produce many identical trees • Needed because the flowers are pollinated by introduced subspecies producing hybrids, so cloning maintains the pure gene pool (but no genetic diversity)	3
04.3	• Stores seeds of wild plants (especially threatened species and crop wild relatives) so species are not lost if extinct in the wild • Seeds collected worldwide; target of 10 000 seeds per species to maintain a diverse gene pool • Stored dry and refrigerated / frozen in an underground reinforced vault to reduce risk of accident • Seeds germinated periodically to check viability and replenish stocks; can be used for reintroduction / breeding	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 35-37; AQA spec 3.1.2.3.3.	
05.1	• Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) • Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) • Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions • Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians)	4
05.2	• Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) • Islands provide roosting / nesting sites safe from terrestrial predators (foxes) • Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks	3
05.3	• Mobile species (birds, flying insects) colonise easily • Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially	2
05.4	• Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
06.1	• Light: selective felling to create clearings for plants needing high light while retaining dense canopy for shade species; clearings also give warm sand / soil for lizard eggs; leave dead wood; woodland floor plants (bluebells) grow before the canopy leaves • Dissolved oxygen: narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae; shallow water warms quickly for egg development • Nutrients: removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground; dung patches add local nutrients	6
06.2	<i>Any 4 from:</i> • Food: wildflower grasslands for seed-eating birds; providing food species • Control of predation: island nest sites unreachable by foxes; trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) • Control of competitors: removing rhododendron so wildflowers get light • Pollination: a range of flowering plants with different flowering times so pollinators have food all season • Seed dispersal species (hippos, elephants, monkeys) conserved; control of introduced pathogens	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
07.1	• A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes	2
07.2	• Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species	4
07.3	• Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass • Choice depends on the vegetation to be controlled and the plants to be protected	2
07.4	• Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground • Heat stimulates dormant heather seeds to germinate so young heather re-establishes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
08.1	• Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump • About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages • Each age has its own community: newly cut areas get high light for wildflowers and butterflies; older areas give dense cover for nesting birds / dormice • Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth	4
08.2	• Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour • Fewer young trees eaten so trees and riverside vegetation recovered • More trees allowed beaver populations to increase; beaver dams increased wetland area • Wetland species (birds, fish, invertebrates) became more common; overall biodiversity rose, though wolves cause problems for livestock farmers	4
08.3	• Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species • Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	